

Question			Marking details	Marks available				Maths	Prac
				AO1	AO2	AO3	Total		
8	(a)	(i)	Use of $\lambda = \frac{hc}{\Delta E}$ including wrong transition or equivalent (1) 1.55 μm (1)		2		2	2	
		(ii)	Absorption: photon disappears or is absorbed or has its energy transferred (1) Electron promoted <u>from L</u> (or ground state) [to U] (1) Don't accept if state going to P Stimulated em: [passing] photon demotes electron <u>from U</u> [to L] (1) [another] photon emitted; 2 photons must be implied (1)	4			4		
		(iii)	Stimulated emission because more electrons <u>in U</u> [than L] Accept population inversion		1		1		
		(iv)	So P stays relatively empty to receive [pumped] electrons or so as not to impede pumping or equivalent or so as to keep the population of U high (1) To maintain high population in U or to maintain population inversion or equivalent (1)	2			2		